COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
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ACTIVITY NO. 1 Algorithm, Flowchart and Pseudocode

1. PROCEDURE OUTPUT

```cpp
#include <iostream>
#include <stack>

int main(){

    std::stack<int> stack1,stack2;
    std::cout << "Testing the stack STL" << std::endl;

    // isEmpty
    std::cout << "Is the stack empty? " << stack1.empty() << std::endl;

    // push
    stack1.push(10);
    std::cout << "The top of the stack is " << stack1.top() << std::endl;
    stack1.push(9);
    std::cout << "The top of the stack is " << stack1.top() << std::endl;
    stack1.push(8);
    std::cout << "The top of the stack is " << stack1.top() << std::endl;
    stack1.push(7);
    std::cout << "The top of the stack is " << stack1.top() << std::endl;

    // pop
    stack1.pop();
    std::cout << "The top of the stack is " << stack1.top() << std::endl;

    std::cout << "Is the stack empty? " << stack1.empty() << std::endl;
    std::cout << "THe size of the stack is " << stack1.size() << std::endl;

    stack1.emplace(11);
    std::cout << "The top of the stack is " << stack1.top() << std::endl;
    std::cout << "THe size of the stack is " << stack1.size() << std::endl;

    std::cout << "The size of the stack is " << std::endl;
    std::cout << "Stack 1: " << stack1.size() << std::endl;
    std::cout << "Stack 2: " << stack2.size() << std::endl;

    stack1.swap(stack2);
    std::cout << "The size of the stack is " << std::endl;
    std::cout << "Stack 1: " << stack1.size() << std::endl;
    std::cout << "Stack 2: " << stack2.size() << std::endl;
    return 0;
}
```

Image 1. Code Snippet for ILO A

This code snippet contains the basic functions for stack such as the implementation of the STL stack. The function push, pop, isEmpty and top. It also contains the implementation of emplace which is basically just adding new data to the array. And swap which switches the two arrays and their data.

```
Testing the stack STL
Is the stack empty? 1
The top of the stack is 10
The top of the stack is 9
The top of the stack is 8
The top of the stack is 7
The top of the stack is 8
Is the stack empty? 0
THe size of the stack is 3
The top of the stack is 11
THe size of the stack is 3
The top of the stack is 11
The top of the stack is 11
THe size of the stack is 4
THe size of the stack is 4
The size of the stack is
The size of the stack is
Stack 1: 4
Stack 1: 4
Stack 2: 0
Stack 2: 0
The size of the stack is
Stack 1: 0
Stack 2: 4
```

Image 2. Output for ILO A

```cpp
#include <iostream>

//global declarations
#define maxCap 100
int stackArr[maxCap];
int top = -1, newData;

//prototype functions
void push();
void pop();
void Top();
bool isEmpty();
bool isFull();
void display();

int main(){
    //main driver code
int choice;
    while(true){
        std::cout << "\n++++++++++++++++++++++++++++++++++++++++++++++++++" << std::endl;
        std::cout << "Stack Operations \n";
        std::cout << "1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display" << std::endl;
        std::cout << "++++++++++++++++++++++++++++++++++++++++++++++++++\n";
        std::cin >> choice;

        switch (choice){
            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: Top();
            break;
            case 4: std::cout << "is stack empty? " << isEmpty() << std::endl;
            break;
            case 5: std::cout << "Is the stack full " << isFull() << std::endl;
            break;
            case 6: display();
            break;
            case 7: return 0;
            break;
            default: std::cout << "Invalid Choice" << std::endl;
        }
    }
}
```

Image 3. Code Snippet for ILO B

This image contains the code snippet for ILO B. It contains the global declarations, prototype functions as well as a switch case format for the main functions which are basically just calling the functions that is defined below depending on what the user demands.

```cpp
bool isEmpty(){
    if(top == -1) return true;
    return false;
}

bool isFull(){
    if (top == maxCap - 1) return true;
    return false;
}
```

Image 4. Code Snippet for ILO B

This basically just checks if the array still has enough room or if it is already full in capacity.

```
void push(){
    if (isFull()){
        std::cout << "Stack Overflow!" << std::endl;
        return;
    }

    std::cout << "Enter a new value: " << std::endl;
    std::cin >> newData;
    stackArr[++top] = newData;
}

void pop(){
    if(isEmpty()){
        std::cout << "Stack Underflow!" << std::endl;
    }

    //display the value that we are going to pop.
    std::cout << "Popping: " << stackArr[top] << std::endl;
    //decrement the top value from the stack
    top--;
}

void Top(){
    //error catching:
    if(isEmpty()){
        std::cout << "The stack is empty." << std::endl;
        return;
    }

    std::cout << "Top Element: " << stackArr[top]<<std::endl;
}
```

Image 5. Code Snippet for ILO B

This image shows the push function which adds a new element. Pop, which removes an element and top which checks for the latest value or data added in the array.

```
void display(){
    if(isEmpty()){
        std::cout << "The stack is empty." << std::endl;
        return;
    }
    std::cout << "This is the content of your stack: ";

    for (int i = top; i >= 0; i--){
        std::cout << stackArr[i];
    }
    std::cout << std::endl;
}
```

Image 6. Code Snippet for ILO B

This is my implementation for the display function of my array. This basically just iterates the whole array and displays the data from the latest entry to the first entry.

```
++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
1
Enter a new value:
1

++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
1
Enter a new value:
2

++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
1
Enter a new value:
3

++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
1
Enter a new value:
4

++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
1
Enter a new value:
5
```

Image 7. Output for Push ILO B

```
++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
2
Popping: 5

++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
3
Top Element: 4

++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
4
is stack empty? 0

++++++++++++++++++++++++++++++++++++++++++++++++
Stack Operations
1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display
++++++++++++++++++++++++++++++++++++++++++++++++
5
Is the stack full 0
```

Image 8. Output for Pop, Top and isEmpty and isFull. ILO B

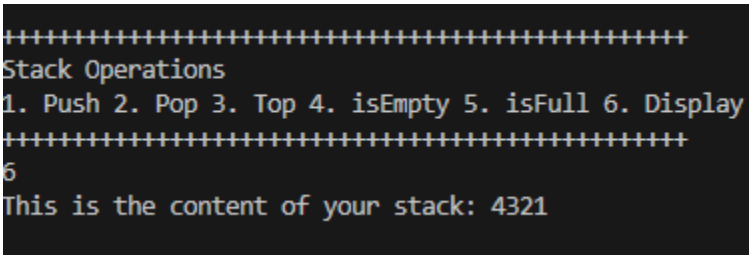


Image 9. Output of display function. ILO B

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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